

JANARDHANA REDDY

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Education

New York University

Sep. 2024 – May 2026

Master of Science in Computer Science

New York City, New York

Relevant Coursework

Software Engineering, Algorithms , Big Data, Machine Learning, Cloud and Big Data, Natural Language Processing

Experience

New York University

Sep 2025 – May 2026

Teaching Assistant, Supervisor: Prof. Sambit Sahu

New York, NY

- Mentored **12+ student teams** on AI-powered capstone projects including **NLP-driven research paper summarization** and **natural language based DevOps automation**, guiding LLM integration, prompt engineering, and RAG pipeline design.
- Taught graduate-level classes on **Kafka, Spark, and NoSQL databases**, and delivered live working demos on distributed data pipelines and cloud-native agentic architectures.
- Led weekly discussions on **latest AI research**, using recent papers on topics including LLM inference optimization, distributed training, and agentic system design for 250 graduate students.

Toyota Kirloskar, Inc

Dec 2023 – Mar 2024

SDE Intern

Bangalore, KA

- Built a logistics mishap dashboard using **Django, PostgreSQL, and REST APIs**, reducing issue resolution time by **30%**.
- Automated invoice approval for a logistics inventory system processing **100,000+** parts and orders, shrinking processing time by **35%** and decreasing error rates by **15%**.
- Enhanced the data processing pipeline by implementing data validation checks, reducing manual effort by **20 hours/week**.

Projects

Project Cerebro: BCI Motor Imagery System | PyTorch, TensorRT, Kafka, PySpark, MLFlow, Go, LangGraph May 2026

- Architected a **distributed BCI pipeline** that reads brain signals non-invasively from paralyzed patients and predicts which direction they intend to move, ingesting **2048 Hz multi channel EEG** via Kafka → MNE → Spark within **500ms latency** across a polyglot store of **Cassandra, MongoDB, Redis & pgvector**.
- Built **cross-hardware transfer learning ensemble** (EEGNet + EEG Conformer + ShallowConvNet, Platt scaled) trained on **140K+ labeled trials**, 162 subjects, achieving **79% macro F1** on held out cross-hardware test data vs **47% baseline**.
- Trained the ensemble across **8 A100 nodes** using **PyTorch DDP** with **torchrun**, batch size 64 per GPU and **NCCL** backend for gradient synchronization, reducing training time to support rapid cross-hardware iteration.
- Optimized PyTorch ensemble inference using **TensorRT FP16** on A100 GPU, reducing inference latency from **250ms to 40–50ms**, an **80%+ reduction** over the baseline FastAPI PyTorch serving pipeline.
- Rewrote the consumer in **Go** after identifying **Python GIL** bottleneck in the Kafka consumer. Used goroutines to enable true parallel execution, dropping latency from **3–5 seconds to sub-500ms**.
- Built an **automated MLFlow evaluation gate** that blocked model promotions failing macro F1 thresholds before deployment, catching **two bad models** that passed single device but failed on unseen device validation.

LeadAI: Agentic GTM Intelligence Platform | LangGraph, FastAPI, Python, PostgreSQL, React, Anthropic API May 2026

- Built a **7 agent LangGraph pipeline** with 5 parallel enrichment agents researching identity, financials, market conditions, and company values across **external APIs**, compressing **45 minutes of manual SDR sales research to under 90 seconds**.
- Implemented a **Claude Sonnet ReAct loop** with native tool calling for autonomous research, dynamically routing API calls based on company type and disqualifying non-qualifying leads in **under 15 seconds** without downstream waste.
- Built a **React and TypeScript frontend** with **SSE streaming** showing a live agent pipeline graph updating in real time, backed by a **FastAPI** backend with **APScheduler** for daily automation and **PostgreSQL** for queue persistence.

Code Sensei | Go, TypeScript, React, AWS EKS, ECS, DynamoDB, LangGraph, PyTorch

May 2025

- Built an **agentic AI powered interview platform** with two specialized LLM agents , an Assessment Agent and a Hint Agent each with **stateless architecture** persisting all session state externally in **DynamoDB and Redis**, enabling any pod to handle any request with full auditability across **1M+ concurrent users** on AWS EKS.
- Implemented **progressive hint level tracking** per question to constrain the hint agent across in three levels from conceptual nudges to pseudocode, combined with **controlled context retrieval** loading only current question, latest code submission, execution result, and prior hints per request, preventing solution leakage while **minimizing prompt token usage**.
- Built **isolated code execution environments** using **ECS Fargate** with CPU and memory caps, network restrictions, and prebuilt language containers, tested adversarially with prompt injection attacks including role bypass and session manipulation until the model consistently redirected users toward reasoning rather than revealing solutions.

Technical Skills

AI & Agents: LangGraph, Anthropic API, OpenAI API, RAG, pgvector, Prompt Engineering, MLFlow, LLM Evaluation

Languages: Python, Go, TypeScript, JavaScript, SQL, C++

Frameworks: FastAPI, Django, React, Node.js, PyTorch, TensorFlow, scikit-learn

Data & Infrastructure: Apache Kafka, Apache Spark, PostgreSQL, Cassandra, MongoDB, Redis, Docker, Kubernetes

Cloud: AWS (EC2, S3, ECS, EKS, Lambda, CloudWatch), Terraform

Concepts: Agentic Architecture, Multi-agent Systems, CI/CD, Distributed Systems, Real-time Pipelines